

Design of Analog Filters for ECG Signal Processing

A project report submitted in partial fulfillment of the requirements for the conferment of the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

by

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July, 2024

CERTIFICATE

This is to certify that the project report titled **Design of Analog Filters for ECG Signal Processing**, submitted by **Ubaidullah (2020BECE009)** and **Nimer Qadir (2020BECE080)** to the Department of Electronics and Communication Engineering at the National Institute of Technology Srinagar, Kashmir, in partial fulfillment of the requirements for the degree of Bachelor of Technology in Electronics and Communication Engineering, is a bonafide record of the project work carried out by him/her under my/our supervision.

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Head of Department

STUDENT DECLARATION

I solemnly affirm that the project report entitled **Design of Analog Filters for ECG Signal Processing**, presented as partial fulfillment for the degree of Bachelor of Technology in Electronics and Communication Engineering, represents original research conducted under the esteemed guidance of **Dr. Pankaj Kumar Sharma**. None of the material contained herein, in whole or in part, has been previously submitted for any academic purpose. I certify that, this work presented is entirely original, free from plagiarism, and has been thoroughly verified for authenticity using Turnitin software.

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Abstract

An electrocardiogram (ECG) quantifies the electrical activity of the heart to screen for different heart diseases, although it can be impacted by noise. ECG signal filtering is a crucial pre-processing step that reduces noise and emphasizes the characteristic waves in ECG data. In real-world applications, the ECG signal is contaminated by different types of noise. Separating the desired signal from noises produced by artifacts such as power line interference (PLI), baseline wandering (BW), and muscular artifacts (MA) is a complicated task.

In this project, we have designed and implemented a 50 Hz operational transconductance amplifier (OTA) based notch filter to attenuate power line interference and an operational transconductance amplifier (OTA)-based low-pass, high-pass and band-pass filter for effectively separating the desired cardiac signal from unwanted noise and artifacts, thereby improving the accuracy and reliability of ECG analysis in clinical and diagnostic applications.

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Chapter 1

Introduction

1.1 Overview

Electrocardiography (ECG) is the graphical recording of the electrical activity of the heart and plays a vital role in the primary diagnosis and monitoring of the health of the heart. The ECG signal is typically in the range of 2mV and requires a recording bandwidth of 0.05 to 100 Hz. Under normal conditions, ECG tracing has very predictable amplitude, direction, and duration. Interpreting these details allows diagnosing a wide range of heart conditions from minor to life-threatening. The ECG signal consists of a P wave, a QRS complex, and a T wave. Figure 1 shows the typical ECG trace.

The electrical activity of the heart is generally sensed by monitoring electrodes placed on the skin's surface. The noises that commonly affect the ECG signal are Power line interference, Baseline Wandering, and Electromyographic Noise (EMG Noise). Power line interference can be easily recognized as the interfering voltage in the ECG may have a frequency of 50 Hz. The most common cause of 50 Hz

interference is the disconnected electrode resulting in a very strong disturbing signal, and therefore needs quick action.

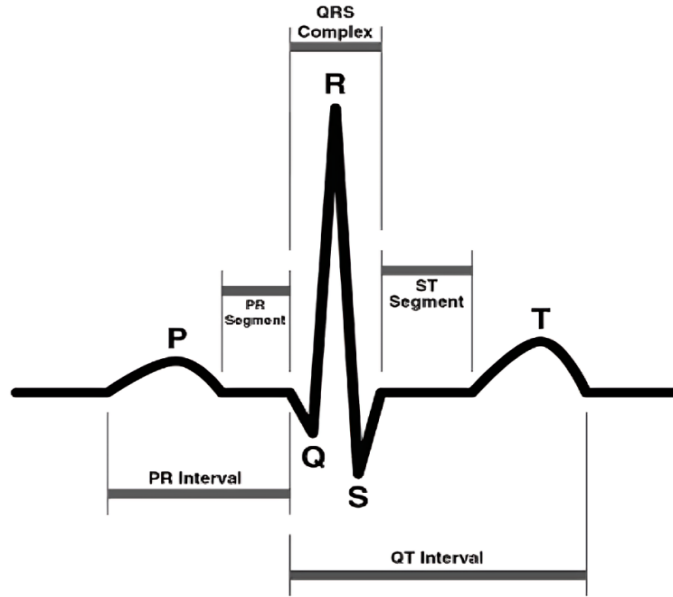


FIGURE 1.1: Typical ECG signal [8]

In the project, we have focused on the design of an operational transconductance amplifier customized to reject power line interference in ECG signals. Further, using this OTA design, we have implemented low-pass, high-pass, and band-pass filters. These filters are designed to help with problems like baseline wandering and electromyographic (EMG) which significantly affect the quality of the ECG signal.

1.2 Problem Statement

To design an analog filter for removing power line interference and optimizing ECG signal processing through band-pass, low-pass, and high-pass configurations with less number of transistors using 180nm technology.

1.3 Practical Applications

1. **Removal of Powerline Interference:** One of the common interferences that an ECG signal can pick up is from the powerline, normally 50 or 60 Hz depending on one's region. When implemented, the notch filter will allow the system to filter out this noisy signal without causing interference with the ECG signal. A power-line filter removes the fuzz in the middle of the ECG signal.
2. **Removal of Baseline Wandering:** Baseline wandering in ECG is often caused due to the patient's breathing, making the signal baseline shift gradually due to poor electrical contact. In most cases, such disturbances occur at frequencies below 0.5 Hz and can be effectively attenuated by using high-pass filters. This way, these filters attenuate most of the frequencies below a certain cutoff value, thereby allowing better accuracy and fidelity in the ECG waveform for diagnostical analysis.
3. **Rejection of Muscle Artifacts:** Muscle artifacts are caused due to muscle movements and may appear as irregular spikes or additional noise on the ECG signal. A low-pass filter efficiently reduces these high-frequency components. Low-pass filters allow lower-frequency signals to pass and block all those above a threshold value. This threshold is usually set to 100 Hz because all of the clinically relevant information falls below this value in the ECG.

1.4 Report Organization

This report is divided into five chapters, the first of which (chapter 1) serves as an introduction to the project work. This chapter covers the necessity for the study, the problem statement, the study's objectives, and a brief synopsis of the report.

The second chapter, (chapter 2) 'Related works', provides a detailed description of prior works in the same topic.

In chapter 3, the design procedure of the Single-stage Operational Transconductance Amplifier (OTA) and the OTA-based filters are explained in detail. This chapter covers the theoretical foundations, design considerations, and the step-by-step process of developing the OTA and the filter.

In Chapter 4, we will see the results obtained from the design of the Single-stage Operational Transconductance Amplifier (OTA) and the OTA-based filters to assess the performance of our design.

In Chapter 5, we will draw conclusions from the results presented in Chapter 4. Chapter 5 will also give the future scope for further research in this area.

Chapter 2

Related Works

2.1 VDCC-based analog comb filter for power line interference cancellation

An analog comb filter is implemented by linking multiple VDCC-based notch filters in a cascading fashion (N in total), eliminating N different pole frequencies. This study focuses on suppressing a fundamental frequency of power-line interference of 50 Hz and its consecutive three odd harmonics at 150 Hz, 250 Hz, and 350 Hz. One significant advantage of this comb filter is the independent control over filters' parameters like quality factor and pole frequency. Additionally, these filters can be electronically tuned by adjusting the transconductance gain of VDCC. The suggested notch filter configuration involves 2 capacitors, 2 resistors, and 1 VDCC element. Extensive simulations were conducted using PSPICE simulator to validate the effectiveness of these filters. The basic building block, VDCC, is designed and implemented in the simulation using integrated circuits MAX435 and AD844.

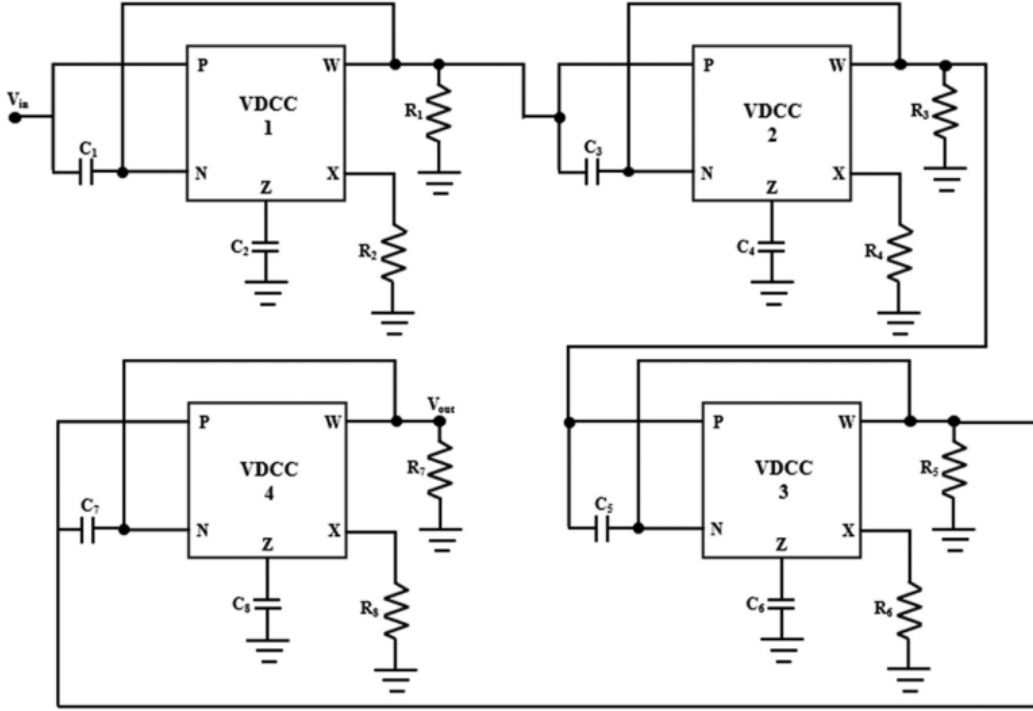


FIGURE 2.1: Comb filter using VDCC-based notch filters [1]

2.2 Power line notch filter using VDGA

This study presents the systematic design and evaluation of a current-mode-based active notch filter aimed at eliminating the pervasive power-line interference (PLI) that corrupts biosignals. The filter is constructed using a modern and versatile current-mode analog building block (ABB), specifically the voltage differencing gain amplifier (VDGA). The design of this second-order filter incorporates a single VDGA as an active component and two capacitors as passive components, eliminating the need for resistors. Employing this filter, the influence of the 50Hz PLI on biological signals is effectively suppressed. The filter exhibits desirable properties of the current-mode circuit, including reduced power consumption, an expanded

dynamic range, and enhanced accuracy. Key parameters of the filter, such as the pole frequency and the quality factor, can be electronically adjusted and orthogonally tuned via the bias currents of the VDGA. To validate its functionality, the proposed filter design was simulated using the PSPICE simulator with the MAX435 IC's macro-model. The simulation results revealed a notch depth and total harmonic distortion (THD) of -52.9 dB and -55 dB, respectively.

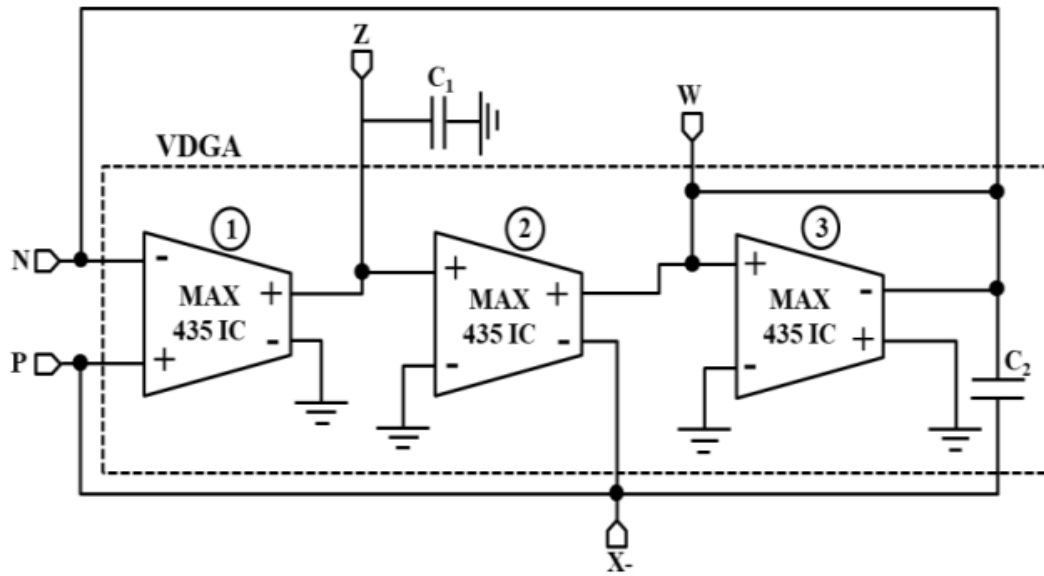


FIGURE 2.2: Realization of power line notch filter using VDGA [2]

2.3 Design of High-pass and Low-pass Filter using CMOS Operational Trans-conductance Amplifier

In this research work, Low-pass and High-pass filters were designed by passive elements using capacitors and resistors and were designed by CMOS transistors.

The Design of OTA was done on Tanner EDA software. The proposed filter consists of OTA, resistors, and capacitors. This filter shows low sensitivity to passive components, low component count, and ease of design. The simulations are done using Tanner EDA version 16.2 at 0.25nm technology.

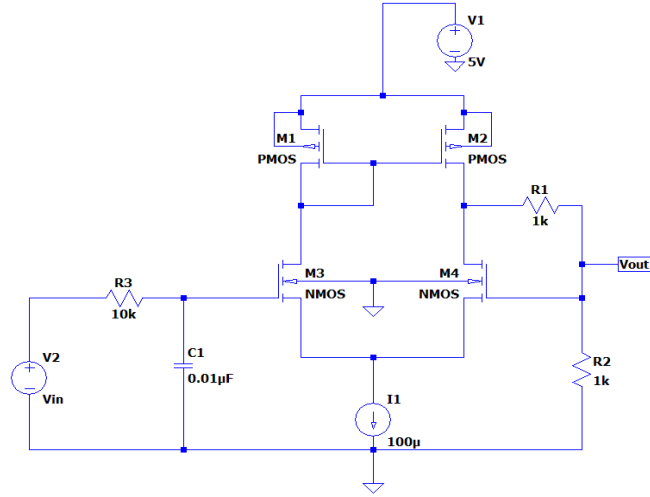


FIGURE 2.3: Schematic of Low-Pass Filter [3]

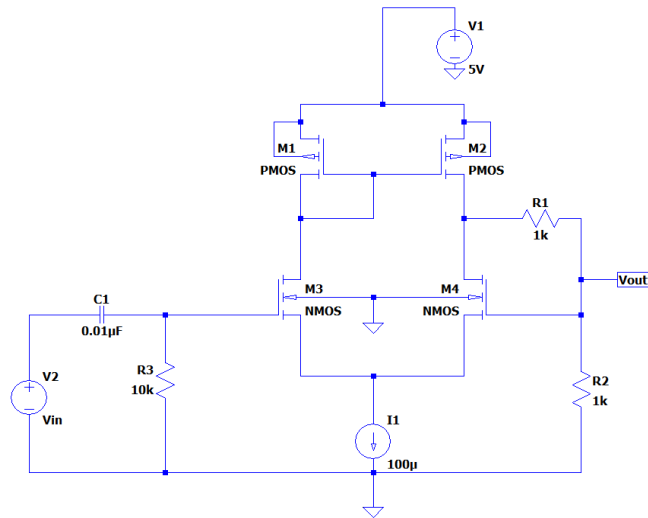


FIGURE 2.4: Schematic of High-Pass filter [3]

Chapter 3

Proposed Work

3.1 Design of Single-Stage OTA

We have used a TSMC 180nm technological file, from there we have got the values of the mobility of holes and electrons, oxide thickness, and threshold voltage of PMOS and NMOS. We have designed our Single-Stage operational transconductance amplifier for the following parameters:

1. $V_{dd} = 1.8V$
2. $C_l = 10pF$
3. $ICMR^+ = 1.6V$
4. $ICMR^- = 0.8V$
5. $P_{dis} \leq 0.3mW$
6. $Slewrate \leq 5V/\mu sec$

7. $A_v = 40db$

8. $UGB \geq 5MHz$

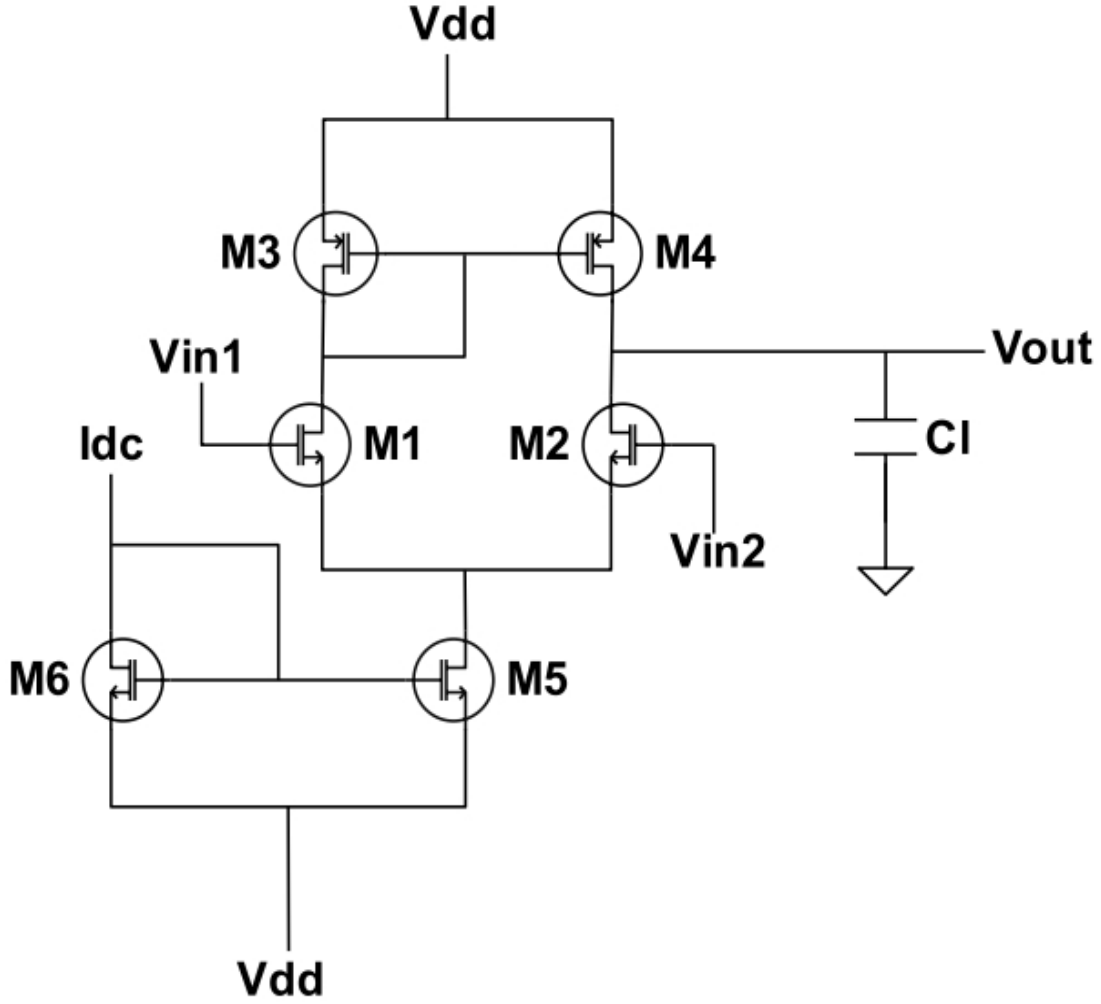


FIGURE 3.1: Schematic of a Single-Stage OTA

3.1.1 Design procedure

1. The OTA's internal capacitance is neglected as the load capacitance is much larger than the lumped parasitic capacitance. First, we are going to find out the value of tail current Id_5 using Slew rate and value of load capacitor C_l .

$$I_{d5} = \text{Slewrate} \times C_l$$

$$I_{d5} = 5V/\mu\text{sec} \times 10pF$$

$$I_{d5} = 50\mu A$$

2. I_{d5} and I_{dc} will be equal as they are forming a current mirror.

$$I_{d5} = I_{dc} = 50\mu A$$

3. I_{d5} divides equally between $M1$ and $M2$ i.e. $I_{d1} = I_{d2} = 25\mu A$

4. For $M1$ transistor to be in saturation

$$V_{DS1} > V_{GS1} - V_{t_n}$$

$$V_D > V_G - V_{t_n}$$

$$V_{t_n} = 0.3662 \text{ (from TSMC 180nm file)}$$

$$V_D > V_{in} - 0.3662 \text{ (Taking } ICMR^+ \text{ value as } V_{in})$$

$$V_D > 1.6 - 0.3662$$

$$V_D > 1.2338$$

$$V_D = V_x = 1.3V$$

5. $Cox = \frac{\epsilon_r \times \epsilon_o}{tox}$

$$Cox = \frac{3.97 \times 8.85 \times 10^{-12}}{4.1 \times 10^{-9}}$$

$$Cox = 8.569 \times 10^{-7} F/cm^2$$

6. $\mu_n = 273.809 cm^2/Vsec$ (from TSMC 180nm file)

7. $\mu_n \times Cox = 234.622 \mu F/Vsec$

8. $\mu_p = 115.689 cm^2/Vsec$ (from TSMC 180nm file)

9. $\mu_p \times Cox = 99.133 \mu F/Vsec$

10. $V_{SG3} = V_S - V_{G3}$

$$V_{SG_3} = V_{dd} - V_x$$

$$V_{SG_3} = 1.8 - 1.3$$

$$V_{SG_3} = 0.5V$$

11. Transistor $M3$ is in saturation because its drain and gate terminal are shorted.

$$I_{d_3} = 25\mu A$$

$$I_{d_3} = \frac{\mu_p C_{ox}}{2} \times \left(\frac{W}{L}\right)_3 \times (V_{SG_3} - |V_{t_p}|)^2$$

$$\left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4 = 42$$

12. $GBP \geq 5MHz$

$$\frac{g_{m_{1,2}}}{2\pi C_l} = 5MHz$$

$$g_{m_{1,2}} = 5MHz \times 2\pi C_l$$

$$g_{m_{1,2}} = 314.16\mu S$$

13. $g_{m_1}^2 = 2 \times I_{d_1} \times \mu_n \times C_{ox} \times \left(\frac{W}{L}\right)_1$

$$\left(\frac{W}{L}\right)_1 = 8.5$$

14. $I_{d_1} = \frac{\mu_n C_{ox}}{2} \times \left(\frac{W}{L}\right)_1 \times (V_{GS_3} - V_{t_n})^2$

$$I_{d_1} = 25\mu A$$

$$V_{GS_3} = 0.524V$$

15. $V_{in_1} = V_{GS_1} + V_{D_{sat}}$

$$0.8 = 0.524 + V_{D_{sat}} \text{ (Taking } I_{CMR}^- \text{ value as } V_{in})$$

$$V_{D_{sat}} = 0.276V$$

16. $I_{d_5} = \frac{\mu_n C_{ox}}{2} \times \left(\frac{W}{L}\right)_5 \times V_{D_{sat}}^2$

$$\left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_6 = 5.6$$

Thus we were able to find $\left(\frac{W}{L}\right)$ ratio of each transistor. The aspect ratios of various transistors are mentioned in the below given table.

MOSFETs	$W(\mu m)$	$L(\mu m)$
M1 and M2	8.5	1
M3 and M4	42	1
M5 and M6	5.6	1

TABLE 3.1: Aspect ratio of MOSFETs

3.2 Design of various analog filters

We have used a second-order OTA-based filter structure to implement low-pass, band-pass, high-pass, and notch filters through the proper selection of the inputs for the circuit shown in Fig 3.2.

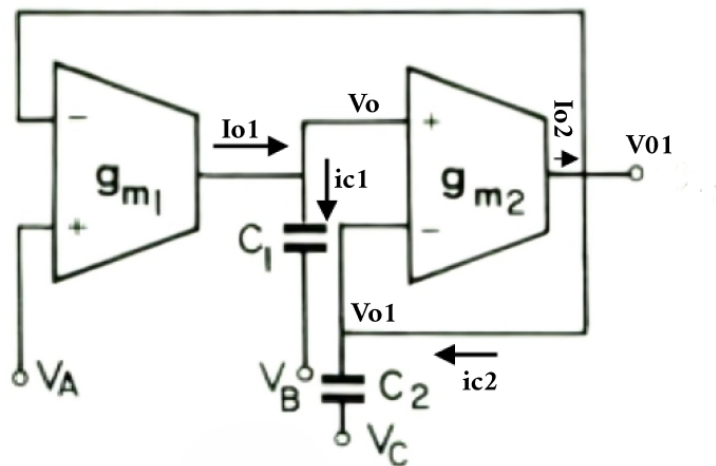


FIGURE 3.2: Second-order filter structure

The analysis of the filter is given by the following steps:

1. $I_{O1} = g_{m1}(V_A - V_{O1}) \dots \dots \dots (1)$

$$2. I_{O1} = i_{C1} = sC_1(V_O - V_B) \dots \dots \dots (2)$$

$$3. I_{O2} = g_{m2}(V_O - V_{O1}) \dots \dots \dots (3)$$

$$4. I_{O2} = i_{C2} = sC_2(V_{O1} - V_C) \dots \dots \dots (4)$$

From (1) & (2)

$$5. g_{m1}V_A - g_{m1}V_{O1} = sC_1(V_O - V_B)$$

$$V_{O1} = \frac{g_{m1}V_A - sC_1V_O + sC_1V_B}{g_{m1}} \dots \dots \dots (5)$$

From (3) & (4)

$$6. g_{m2}(V_O - V_{O1}) = sC_1(V_O - V_B)$$

$$V_O = \frac{sC_2V_{O1} - sC_2V_C + g_{m2}V_{O1}}{g_{m2}} \dots \dots \dots (6)$$

Put (6) in (5)

$$7. V_{O1} = \frac{g_{m1}g_{m2}V_A + sC_1g_{m2}V_B + s^2C_1C_2V_C}{g_{m1}g_{m2} + s^2C_1C_2 + sC_1g_{m2}} \dots \dots \dots (7)$$

The last equation represents the output of the filter.

The transfer function for the specific inputs at V_A , V_B , and V_C are listed in the Table for lowpass, bandpass, highpass, and notch versions of this circuit

Circuit Type	Input Conditions	Transfer function
Low-pass	$V_i = V_A, V_B$ and V_C Grounded	$\frac{g_{m1}g_{m2}}{s^2C_1C_2 + sC_1g_{m2} + g_{m1}g_{m2}}$
High-pass	$V_i = V_C, V_B$ and V_C Grounded	$\frac{s^2C_1C_2}{s^2C_1C_2 + sC_1g_{m2} + g_{m1}g_{m2}}$
Band-pass	$V_i = V_B, V_A$ and V_C Grounded	$\frac{sC_1g_{m2}}{s^2C_1C_2 + sC_1g_{m2} + g_{m1}g_{m2}}$
Notch	$V_i = V_A = V_C$ and V_B Grounded	$\frac{s^2C_1C_2 + g_{m1}g_{m2}}{s^2C_1C_2 + sC_1g_{m2} + g_{m1}g_{m2}}$

TABLE 3.2: Transfer function for various filter configurations

3.2.1 Design of 50 Hz Notch Filter

We are going to design a 50Hz Notch filter that is going to attenuate the power line interference. The notch filter will allow the system to filter out the power line interference without causing interference with the ECG signal.

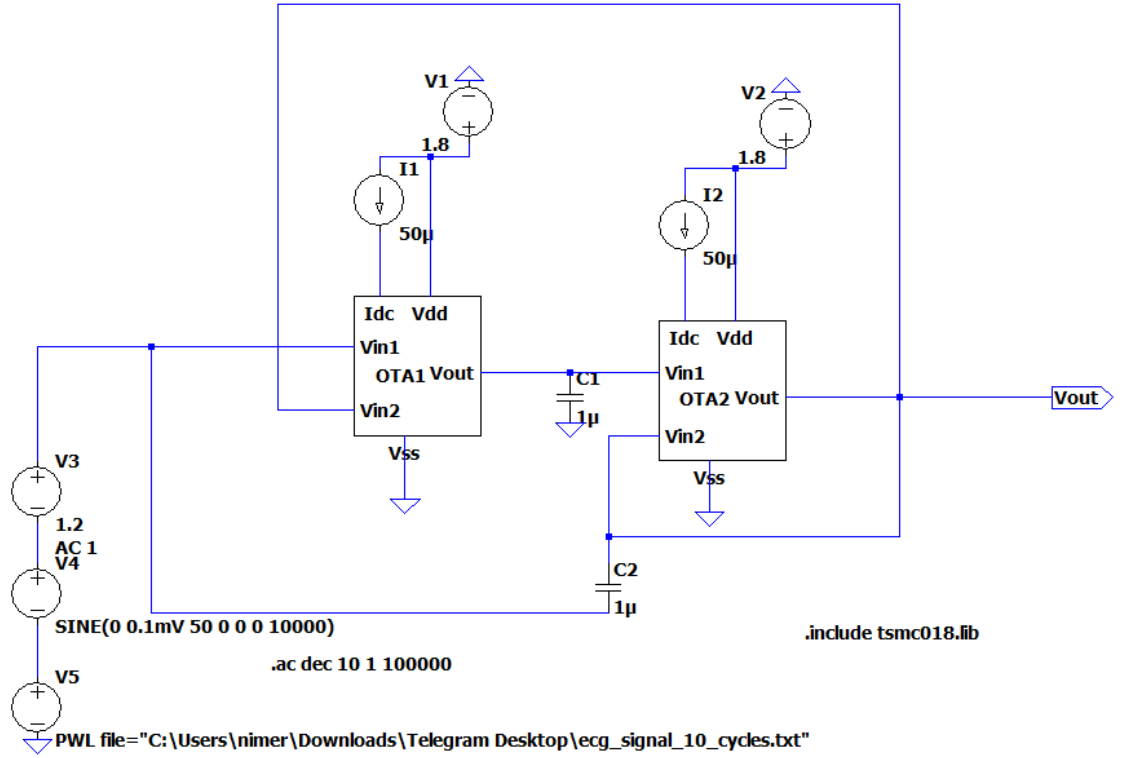


FIGURE 3.3: Schematic of an OTA-based 50 Hz Notch filter

3.2.1.1 Design parameters

1. Transfer function = $\frac{g_{m1} g_{m2} V_A + s^2 C_1 C_2 V_C + s C_1 g_{m2} V_B}{g_{m1} g_{m2} + s^2 C_1 C_2 + s C_1 g_{m2}}$
2. $g_{m1} = g_{m2} = g_m = 314.16 \mu S$
3. Notch Frequency = $f_o = \frac{1}{2\pi} \times \frac{g_m}{\sqrt{C_1 C_2}}$

4. On solving and doing trial and error we came to find the value of capacitors

$$C_1 \text{ and } C_2 \quad C_1 = C_2 = 1\mu F$$

5. On using these capacitor values we get $f_o = 50.025Hz$ (theoretical)

$$6. Q = \sqrt{\frac{C_2}{C_1}} = 1$$

3.2.2 Design of Low-Pass Filter

We are going to design a Low-Pass filter having a cutoff frequency of 100 Hz to prevent muscle artifacts.

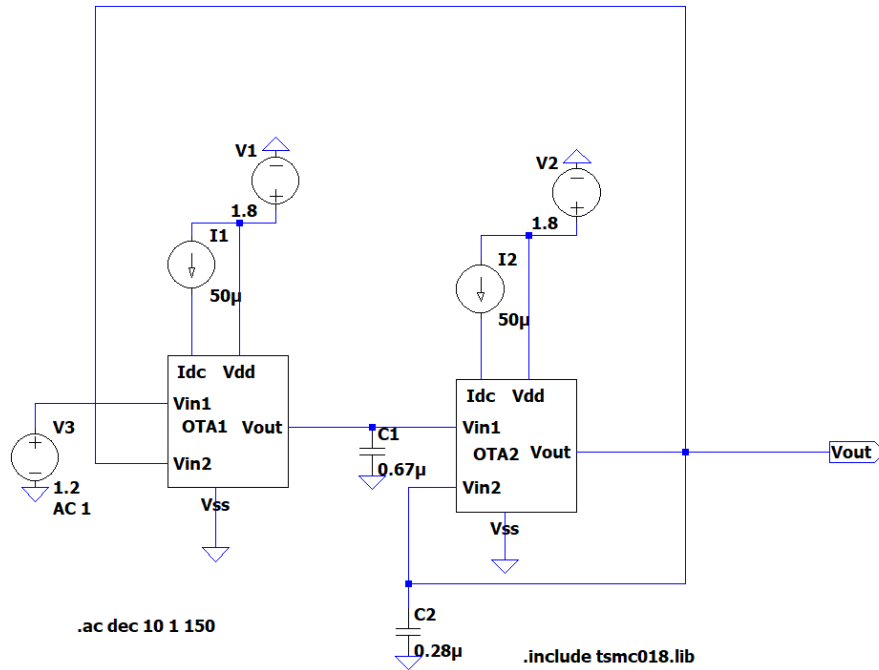


FIGURE 3.4: Schematic of an OTA-based Low-Pass Filter

3.2.2.1 Design parameters

$$1. \text{ Transfer function} = \frac{g_{m1}g_{m2}}{s^2C_1C_2 + sC_1g_{m2} + g_{m1}g_{m2}}$$

$$2. g_{m1} = g_{m2} = g_m = 314.16\mu S$$

3. Cutoff Frequency = $f_c = \frac{1}{2\pi} \times \frac{g_m}{\sqrt{C_1 C_2}}$
4. On solving and doing trial and error we came to find the value of capacitors C_1 and C_2
 $C_1 = 0.67\mu$ and $C_2 = 0.28\mu F$
5. On using these capacitor values we get $f_c = 115.439Hz$ (theoretical)
6. $Q = \sqrt{\frac{C_2}{C_1}} = 0.64$

3.2.3 Design of High-Pass Filter

We are going to design a High-pass filter having a cutoff frequency of 0.05 Hz to prevent Baseline Wandering.

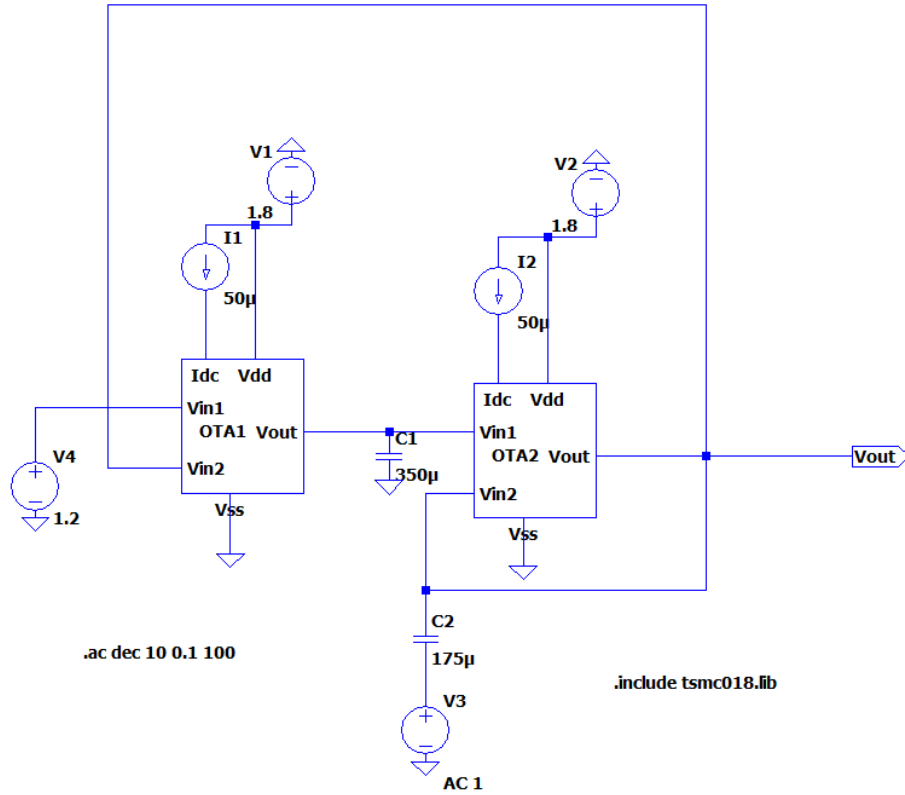


FIGURE 3.5: Schematic of an OTA-based High-Pass Filter

3.2.3.1 Design parameters

1. Transfer function = $\frac{s^2 C_1 C_2}{g_{m1} g_{m2} + s^2 C_1 C_2 + s C_1 g_{m2}}$
2. Cutoff Frequency = $f_c = \frac{1}{2\pi} \times \frac{g_m}{\sqrt{C_1 C_2}}$
3. On solving and doing trial and error we came to find the value of capacitors C_1 and C_2
 $C_1 = 350\mu$ and $C_2 = 175\mu F$
4. On using these capacitor values we get $f_c = 0.202Hz$ (theoretical)
5. $Q = \sqrt{\frac{C_2}{C_1}} = 0.70$

3.2.4 Design of Band-Pass Filter

We are going to design a band pass filter having lower cutoff frequency of 0.2Hz and an upper cutoff frequency of approx 100 Hz.

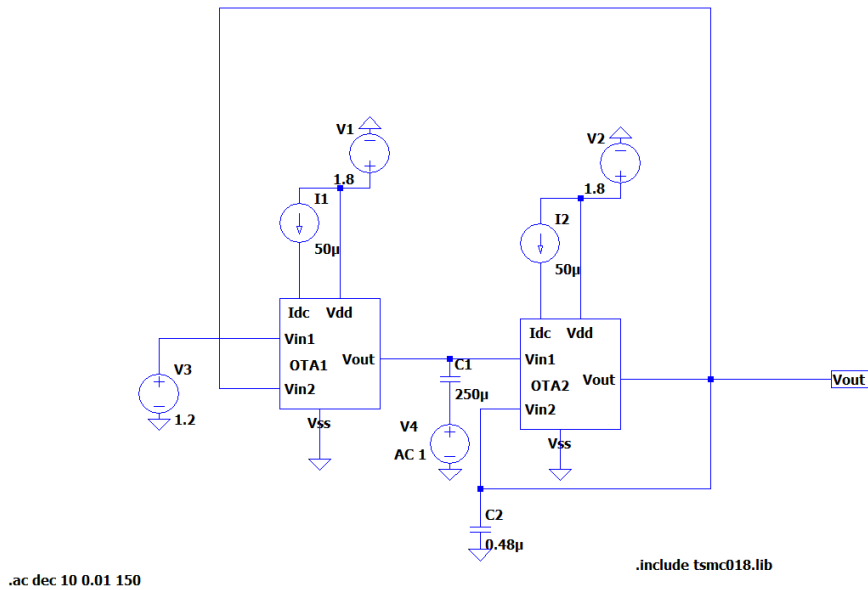


FIGURE 3.6: Schematic of an OTA-based Band-pass filter

3.2.4.1 Design parameters

1. Transfer function = $\frac{sC_1g_{m2}}{g_{m1}g_{m2} + s^2C_1C_2 + sC_1g_{m2}}$
2. $g_{m1} = g_{m2} = g_m = 314.16\mu S$
3. Centre Frequency = $f_o = \frac{1}{2\pi} \times \frac{g_m}{\sqrt{C_1C_2}}$
4. On solving and doing trial and error we came to find the value of capacitors C_1 and C_2
 $C_1 = 250\mu F$ and $C_2 = 0.48\mu F$
5. On using these capacitor values we get $f_o = 4.566Hz$ (theoretical)
6. $Q = \sqrt{\frac{C_2}{C_1}} = 0.0438$

Chapter 4

Results and discussions

4.1 Simulation Results of Single-Stage OTA

We applied a sinusoidal differential input of amplitude $4mV$ at the input terminals and got an amplified version of the input at the output terminal. We have used a $V_{dd} = 1.8V$, $V_{ss} = 0V$ and $I_{dc} = 50\mu A$ for which we got the following results, and all the simulations are performed in LT SPICE XVII software.

4.1.1 Differential input waveform

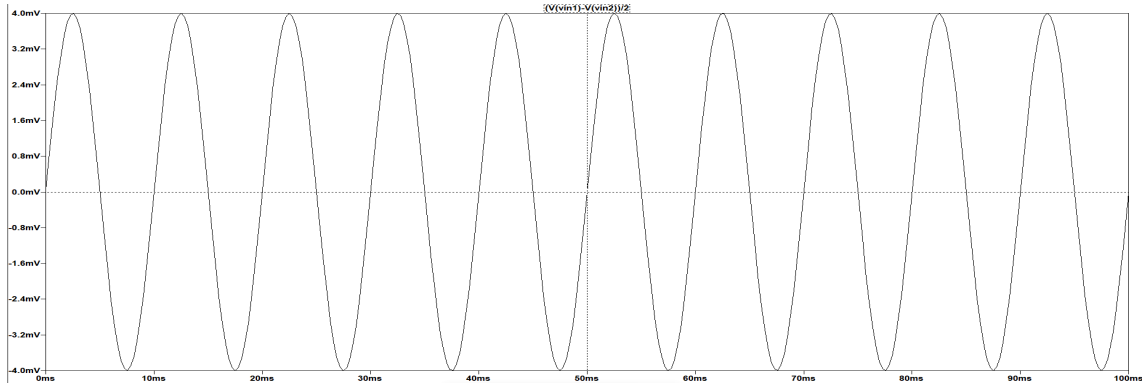


FIGURE 4.1: Differential input waveform

4.1.2 Output waveform

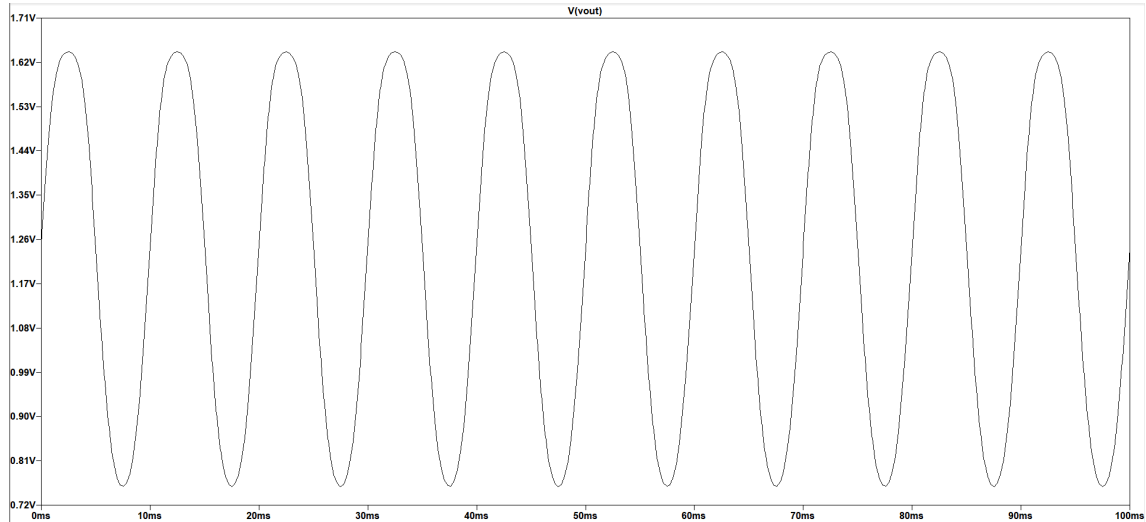


FIGURE 4.2: Output waveform

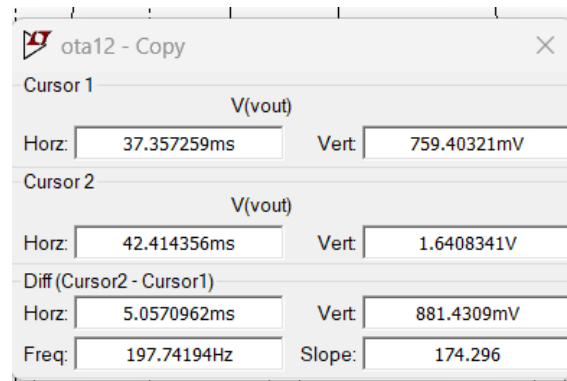


FIGURE 4.3: Reading of Peak to peak output voltage reading

4.1.2.1 Observation

Differential input voltage had peak amplitude = 4mV

Output waveform had peak to peak amplitude=881.4309mV

So Output waveform had peak amplitude=440.715mV

$$\text{Gain} = \frac{440.715}{4} = 110.17$$

So it is evident from the above results that our OTA was successfully able to amplify the differential input signal.

4.1.3 Frequency response of Single-stage OTA

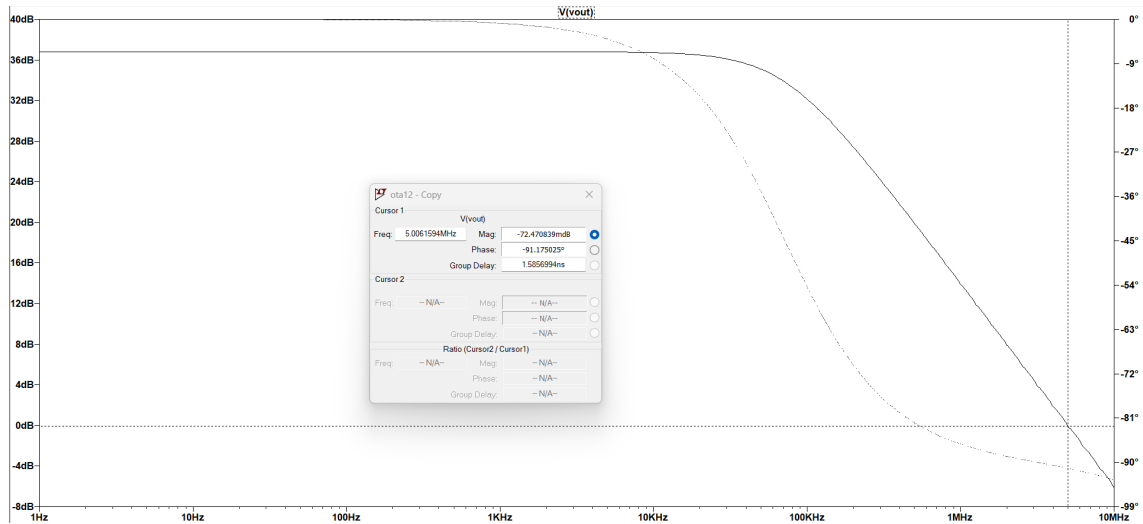


FIGURE 4.4: Frequency response of Single-stage OTA

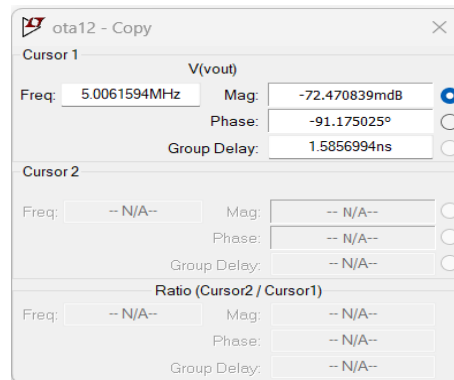


FIGURE 4.5: Reading of Frequency response of Single-stage OTA

4.1.3.1 Observation

The frequency response cuts the 0-db line at 5MHz therefore we are successfully able to get a gain-bandwidth product of 5MHz.

4.2 Simulation Results of Notch Filter

We begin by generating a clean ECG signal, then we superimpose it with 50 Hz noise, and then we apply the noisy ECG signal to our notch filter. Fig 4.6 shows the clean ECG signal (without noise), Fig 4.7 shows the ECG signal with noise, and Fig 4.8 shows the ECG signal at the output of our notch filter.

4.2.1 ECG signal without 50Hz noise

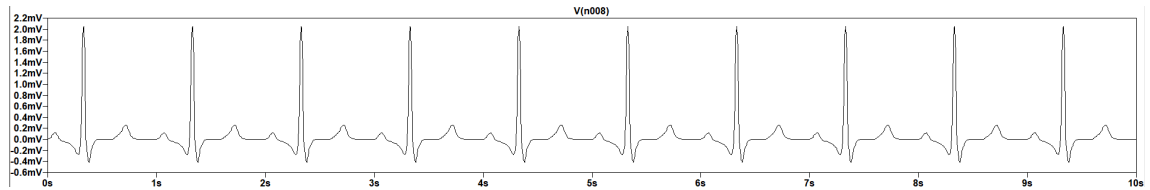


FIGURE 4.6: ECG signal without 50Hz noise

4.2.2 ECG signal with 50Hz noise

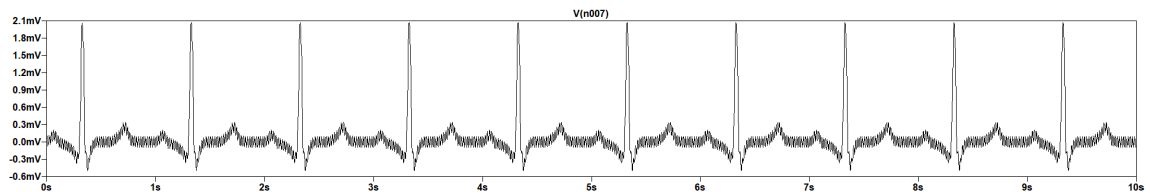


FIGURE 4.7: ECG signal with 50Hz noise

4.2.3 ECG signal at the output of notch filter

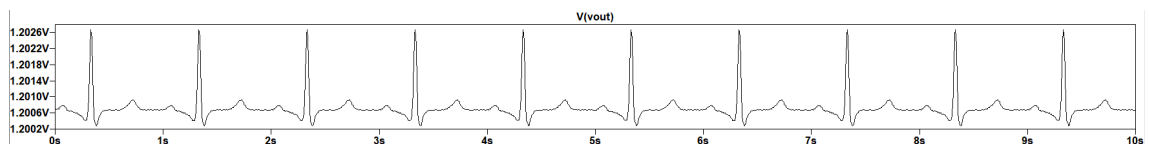


FIGURE 4.8: ECG signal at output of notch filter

4.2.4 Observation

We were successfully able to remove 50Hz noise using our OTA-based notch filter.

4.2.5 Frequency response of Notch filter

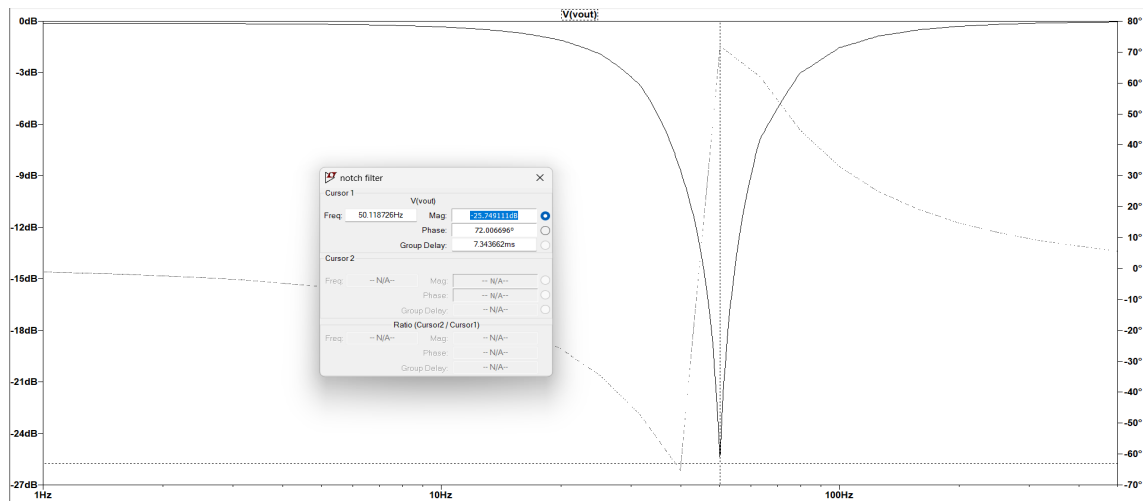


FIGURE 4.9: Frequency response of Notch filter

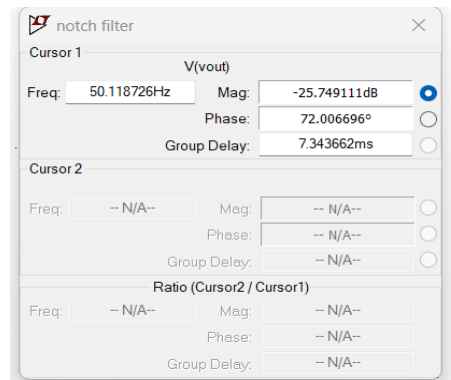


FIGURE 4.10: Reading of Frequency response of Notch filter

4.2.5.1 Observation

We got a notch at 50.118Hz frequency with a notch depth equal to -25.74911dB.

4.3 Simulation Results of other filters

Now we will analyze the frequency response of low-pass, high-pass, and band-pass filters and observe their cutoff frequencies.

4.3.1 Frequency response of Low-pass filter

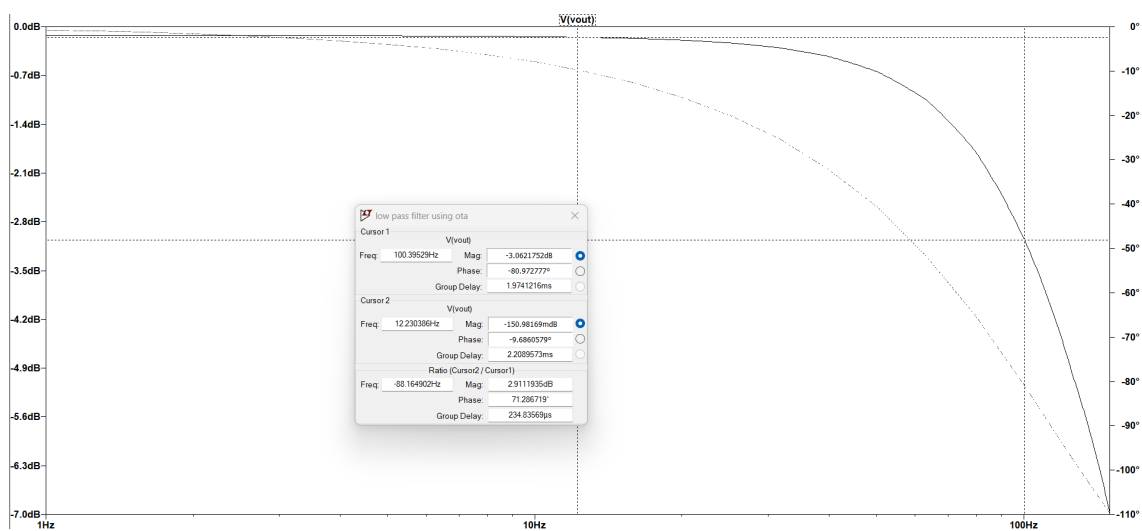


FIGURE 4.11: Frequency response of Low-pass filter

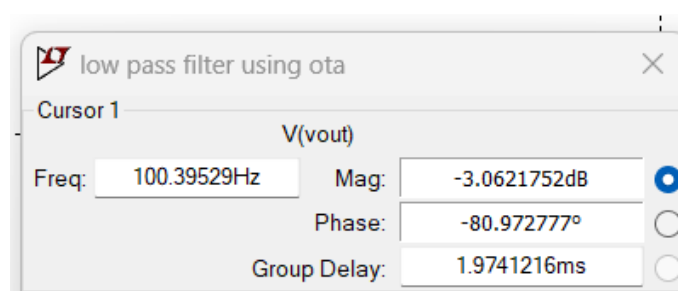


FIGURE 4.12: Reading of Frequency response of Low-pass filter

4.3.1.1 Observation

We were able to design a low-pass filter having a cutoff frequency of 100.39Hz.

4.3.2 Frequency response of High-pass filter

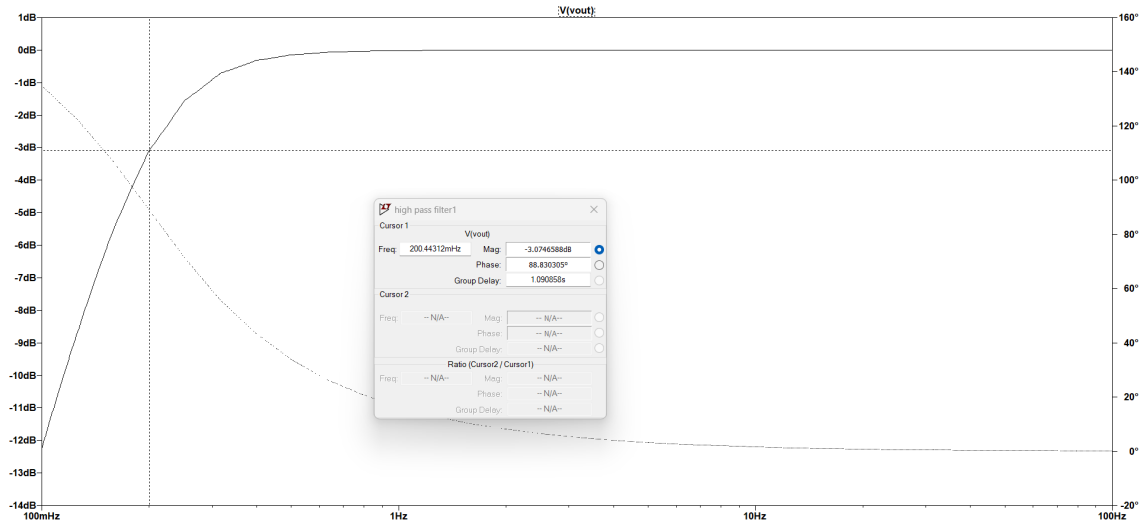


FIGURE 4.13: Frequency response of High-pass filter

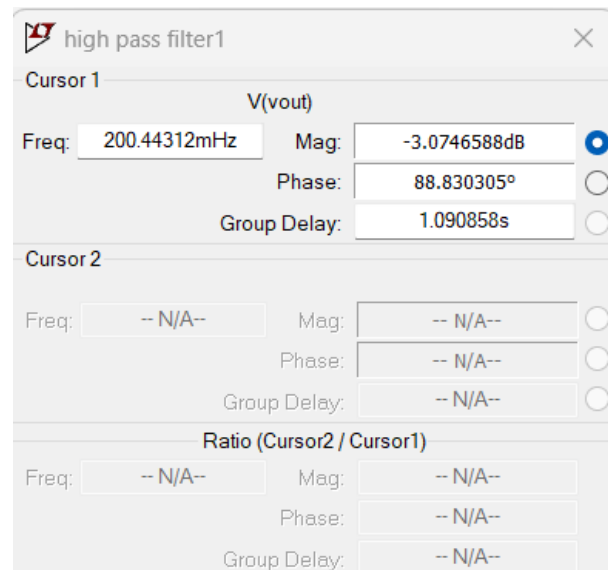


FIGURE 4.14: Reading of Frequency response of High-pass filter

4.3.2.1 Observation

We were able to design a High-pass filter having a cutoff frequency of 200.44mHz.

4.3.3 Frequency response of Band-pass filter

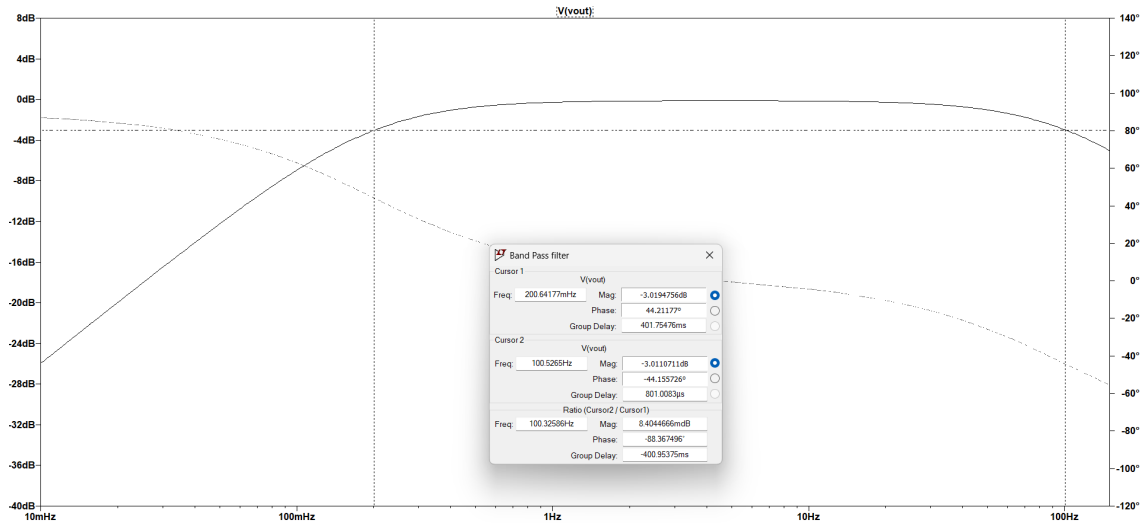


FIGURE 4.15: Frequency response of Band-pass filter

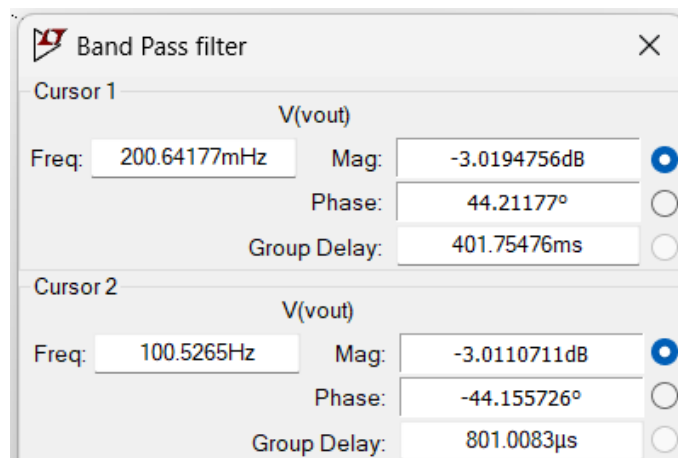


FIGURE 4.16: Reading of Frequency response of Band-pass filter

4.3.3.1 Observation

We were able to design a Band-pass filter having a lower cutoff frequency of 200.64mHz and an upper cutoff frequency of 100.52Hz.

Chapter 5

Conclusion and Future scope

5.1 Conclusion

In this study, we have addressed the challenging task of removing ECG interferences like power line interference, baseline wandering, and EMG noise, which normally results in improper diagnosis and sometimes even in life-threatening situations. We have proposed a notch filter to address power line interference, a high pass filter for baseline wandering, and a low pass filter for EMG noise. Besides that, we came up with a band-pass filter design for the prevention of baseline wandering and EMG noise. These designs have proven effective in significantly improving the quality of ECG signals, facilitating better detection and diagnosis.

5.2 Future scope

This design can further be improved by designing higher-order filters which can significantly improve the accuracy and performance of signal processing.

Higher-order filters can provide sharper cutoff frequencies and better attenuation of unwanted signals, leading to clearer and more accurate ECG readings.

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